NATIONAL MATHEMATICS CONTEST
PRIMARY-LEVEL PAPER 1.

NAMES: _____ DATE OF BIRTH: _____

SCHOOL/INSTITUTION: _____

CLASS: _____ DISTRICT: _____

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

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PLEASE, NOTE CAREFULLY:

1. **WRITE YOUR ANSWERS IN THE SPACES PROVIDED. MORE SPACE IS AVAILABLE AT THE BACK OF EACH PAGE.**
2. Date: Saturday, 17th March, 2018.
3. Time: Exactly 2 hours and 15 Minutes (from 9 : 00am to 11 : 15am)
4. Type:
   (a) This paper consists of two sections and each section carries 50 marks.
   (b) Attempt as many questions as you can. The total marks scored from all the questions you attempt will be your final score.
   (c) No marks will be awarded for an answer if a clear and logical layout of the working is not shown.
5. All diagrams/figures in this contest are not to scale.
6. Immediately the paper is done, the answer script(s) must be forwarded to:

   David Ddumba Walakira,
   Contest Coordinator, Uganda Mathematical Society,
   c/o Department of Mathematics, Makerere University,
   P.O. Box 7062, Kampala.
   Telephone contact: $+256-414-540692$.

   Answer scripts should be received **latest, Monday** 19**th March,** 2018.
7. **ALL** *participants who qualify for Paper* 2 *are cordially invited to the certificate and prize giving ceremony on Saturday* 28*th July,* 2018 *at NOON at Makerere University, Kampala.*
8. National Mathematics Contest 2018 Paper 2 will be done on 23rd June, 2018 at various centres.
9. **UGANDA MATHEMATICAL SOCIETY** wishes you success in this year's contest.

SECTION A : 5 marks each

1. Five goats marked A, B, C, D, E took part in a race. The time, in seconds, taken by each goat to complete the race is as given below. Which goat won the race?(*Circle the right answer*).
   $A:$ 2.011 $B:$ 2.101 $C:$ 2.010 $D:$ 2.001 $E:$ 2.200

2. A number line has 100 consecutive integers marked on it. If the smallest of these integers is -13, what is the largest?

3. Rectangular tiles, measuring $6cm \times 4cm$ are arranged without overlapping, to create a square. What is the minimum number of tiles needed to make a square?

4. If $\frac{1}{2}$ of $\frac{2}{5}$ of the 15 small squares in the figure are removed, how many squares remain?



5. The first six letters of the alphabet A, B, C, D, E, F are assigned values $1, 2, 3, 4, 5, 6$ respectively. The value of a word equals the sum of the values of its letters, for example, BEEF is $18 = 2+5+5+6$. Which of the following words has the biggest value: FADE, BEAD, FEED, FACE, DEAF?

6. In the following equations, the letters a, b, c represent different numbers. What is the numerical value of $a + b + c$?

$$\begin{aligned} 1^3 &= 1 \\ a^3 &= 1+7 \\ 3^3 &= 1+7+b \\ 4^3 &= 1+7+c \end{aligned}$$

7. Sylvia sat a mathematics examination paper that had 30 questions in Section A and 50 questions in Section B. Each question in each section was worth 1 mark. She got 70% of the Section A questions correct and her overall examination mark was 80%. How many Section B questions did Sylvia answer correctly?

8. A Prime Movie Hall has 13 rows numbered from 1 to 13. Each row numbered with a prime number has 17 seats. The rest of the rows have 15 seats each. How many seats are there in the Prime Movie Hall?

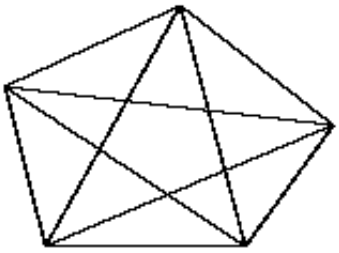
9. Allen is 6 years older than Jane. Jane is 3 years younger than Doreen. If Allen is 19 years old, how old is Doreen?

10. The Head teacher takes one hour to drive the 59km from her home to school. Her drive includes 20 minutes on a highway and 40 minutes on city roads. If her average speed when she is on city roads is 45km/h, what is her average speed on the highway in km/h?

SECTION B : 10 **marks each**

11. A primary school has two primary seven streams (P7W and P7S). In P7W, the ratio of girls to boys is 2 : 3. In P7S, the ratio of girls to boys is 3 : 5. If there are 20 girls in both streams, what is the least number of boys in both streams?

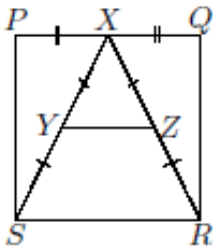
12. How many triangles are in the figure?



13. Cathy runs 100 metres in 10 seconds. Sarah runs 400 metres in 44 seconds. They both participate in a 1000 metre race; each maintaining her speed. How far ahead, to the nearest metre, is the winner as she crosses the finishing line?

14. At some hospital with several operating rooms and several doctors, twenty people are each waiting for a 45-minute operation. The first operation starts at 8.00am, the second operation at 8.15am, and each of the rest of operation starts at 15 minute intervals thereafter. When does the last operation end?

15. In the figure below, $PQRS$ is a square with side length 8 cm. X is the midpoint of side PQ and Y and Z are midpoints of XS and XR, respectively. What is the area of $YZRS$?



NATIONAL MATHEMATICS CONTEST
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SECTION A : 5 marks each

1. Five goats marked A, B, C, D, E took part in a race. The time, in seconds, taken by each goat to complete the race is as given below. Which goat won the race?(*Circle the right answer*).
   $A:$ 2.011 $B:$ 2.101 $C:$ 2.010 $D:$ 2.001 $E:$ 2.200

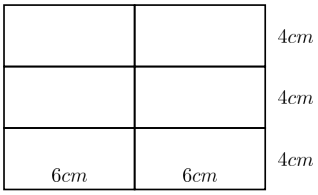
   Solution 1 *The goat with the least time taken, won the race, this is goat "D".*

2. A number line has 100 consecutive integers marked on it. If the smallest of these integers is -13, what is the largest?

   Solution 2

$$\begin{aligned} \textit{The largest integer} &= \textit{Starting point} + \textit{No. of integers} - 1 \\ &= -13 + 100 - 1 \\ &= 100 - 14 \\ &= 86 \end{aligned}$$

3. Rectangular tiles, measuring $6cm \times 4cm$ are arranged without overlapping, to create a square. What is the minimum number of tiles needed to make a square?



   Solution 3 6×4 *lengths are in* $3:2$*, so you need* $2 \times 3 = 6$ *tiles.*

4. If $\frac{1}{2}$ of $\frac{2}{5}$ of the 15 small squares in the figure are removed, how many squares remain?



   Solution 4

$$\begin{aligned} \textit{Numbers of squares removed} &= \frac{1}{2} \times \frac{2}{5} \times 15 \\ &= \frac{1}{15} \times 15 \\ &= 3 \end{aligned}$$

   Number of squares remaining is $15 - 3 = 12$ *squares.*

5. The first six letters of the alphabet A, B, C, D, E, F are assigned values $1, 2, 3, 4, 5, 6$ respectively. The value of a word equals the sum of the values of its letters, for example, BEEF is $18 = 2 + 5 + 5 + 6$. Which of the following words has the biggest value: FADE, BEAD, FEED, FACE, DEAF?

Solution 5

$$\begin{aligned} FADE &= 6 + 1 + 4 + 5 = 16 \\ BEAD &= 2 + 5 + 1 + 4 = 12 \\ FEED &= 6 + 5 + 5 + 4 = 20 \\ FACE &= 6 + 1 + 3 + 5 = 15 \\ DEAF &= 4 + 5 + 1 + 6 = 16 \end{aligned}$$

Therefore the answer is FEED

6. In the following equations, the letters a, b, c represent different numbers. What is the numerical value of $a + b + c$?

$$\begin{aligned} 1^3 &= 1 \\ a^3 &= 1 + 7 \\ 3^3 &= 1 + 7 + b \\ 4^3 &= 1 + 7 + c \end{aligned}$$

Solution 6 *From* $a^3 = 1 + 7 = 8, \Rightarrow a = \sqrt[3]{28} = 2$. *Therefore*

$$\begin{aligned} 3^3 &= 1 + 7 + b \\ 27 &= 8 + b \end{aligned}$$

To have $b = 27 - 8 = 19$. *Also*

$$\begin{aligned} 4^3 &= 1 + 7 + c \\ 64 &= 8 + c \end{aligned}$$

To have $c = 64 - 8 = 56$. *Therefore* $a + b + c = 2 + 19 + 56 = 77$

7. Sylvia sat a mathematics examination paper that had 30 questions in Section A and 50 questions in Section B. Each question in each section was worth 1 mark. She got 70% of the Section A questions correct and her overall examination mark was 80%. How many Section B questions did Sylvia answer correctly?

Solution 7 *Overall mark of* 80% *means*

$$\textit{Sylvia got } 80\% \times 80 = \frac{80}{100} \times 80 = 64 \textit{ questions correct}$$

Sylvia got 70% *of section A questions correct, which imply, Section A questions answered correctly* $70\% \times 30 = \dfrac{70}{100} \times 30 = 21$ *questions.*

Therefore, the number of section B questions answered correctly are $64 - 21 = 43$.

8. A Prime Movie Hall has 13 rows numbered from 1 to 13. Each row numbered with a prime number has 17 seats. The rest of the rows have 15 seats each. How many seats are there in the Prime Movie Hall?

Solution 8 *There are* 6 *prime numbers between* 1 *and* 13 *inclusive, namely* $2, 3, 5, 7, 11$ *and* 13. *And* 7 *non-prime numbers. Therefore, the number of seats are*

$$\begin{aligned} &= (17 \times 6) + (15 \times 7) \\ &= 102 + 105 \\ &= 207 \end{aligned}$$

9. Allen is 6 years older than Jane. Jane is 3 years younger than Doreen. If Allen is 19 years old, how old is Doreen?

Solution 9 *Let Jane's age be* x *years. Then*

$$\begin{aligned} &\textit{Allen is } x + 6 \textit{ years old} \\ &\textit{Doreen is } x + 3 \textit{ years old} \end{aligned}$$

If Allen is 19 *years old, then* $x + 6 = 19 \Rightarrow x = 19 - 6 = 13$. *Therefore, Doreen is* $x + 3 = 13 + 3 = 16$ *years old.*

10. The Head teacher takes one hour to drive the 59km from her home to school. Her drive includes 20 minutes on a highway and 40 minutes on city roads. If her average speed when she is on city roads is 45km/h, what is her average speed on the highway in km/h?

Solution 10 *In* 40 *min at* 45 Km/h, *she covers* 30 Km $\left(\frac{2}{3}hr \times 45\ Km/h\right)$. *On highway, she covers* $59 - 30 = 29\ Km$, *in* 20 *mins* $\left(\frac{1}{3}\ hrs\right)$, *to have the average speed on highway as* $\left(\frac{29\ Km}{\frac{1}{3}\ hrs} = 87\ Km/h\right)$

SECTION B : 10 **marks each**

11. A primary school has two primary seven streams (P7W and P7S). In P7W, the ratio of girls to boys is 2 : 3. In P7S, the ratio of girls to boys is 3 : 5. If there are 20 girls in both streams, then what is the least number of boys in both streams?

Solution 11

7W		*S*
G:B	*G:B*	
a students	*b students*	

Therefore

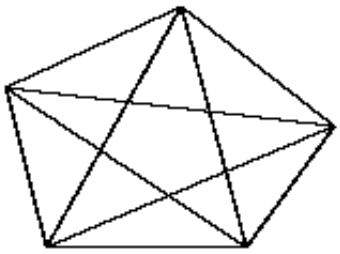
$2a$ *girls and* $3a$ *boys*
$3b$ *girls and* $5b$ *boys*

$$\begin{aligned} 2a + 3b &= 20 \\ 3a + 5b &= ? \end{aligned}$$

2a+b	*a*	*b*	*3a+5b*
20	*1*	*6*	*33*
20	*4*	*4*	*32*
20	*7*	*2*	*31*

Others give non-whole numbers, so 31.

12. How many triangles are in the figure?



Solution 12 *The answer is* 35

35 : 10 *marks*
34 : 9 *marks*
33 : 8 *marks*
32 : 7 *marks*
31 : 6 *marks*
30 : 5 *marks*
29 : 4 *marks*
28 : 3 *marks*
27 : 2 *marks*
26 : 1 *marks*

Any other solution, zero marks.

13. Cathy runs 100 metres in 10 seconds. Sarah runs 400 metres in 44 seconds. They both participate in a 1000 metre race; each maintaining her speed. How far ahead, to the nearest metre, is the winner as she crosses the finishing line?

Solution 13 *Sarah runs* 400 *meters in* 44 *seconds, that it,* 100 *meters in* 11 *seconds. So Cathy runs faster than Sarah. If Cathy runs* 100 *meters in* 10 *seconds, she will complete in* $\frac{1000}{100} \times 10 = 100$ *seconds.*

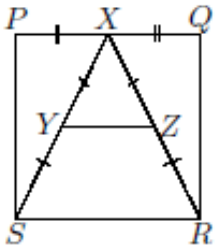
In 100 *seconds, Sarah would have run only* $\frac{400}{44} \times 100 = 909.1$ *meters. Therefore, it will be approximately* $1000 - 909.1 = 90.9$ *meters or* 91 *meters.*

14. At some hospital with several operating rooms and several doctors, twenty people are each waiting for a 45-minute operation. The first operation starts at 8.00am, the second operation at 8.15am, and each of the rest of operation starts at 15 minute intervals thereafter. When does the last operation end?

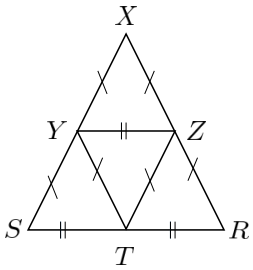
Solution 14 *Since there are* 20 *people scheduled and each begins* 15 *minutes after previous one, the* 20th *operation will begin* $19^{th} - 15$ *minute intervals after the first operation.*

Since began at $8:00am$, $19^{th} - 15$ *minute intervals is* 285 *minutes, that is,* 4 *hours* 45 *minutes* $20^{th} - 45$ *minute operation begins at* $12:45$ *and ends* $12:45 + 45\ min = 1:30\ pm$.

15. In the figure below, $PQRS$ is a square with side length 8 cm. X is the midpoint of side PQ and Y and Z are midpoints of XS and XR, respectively. What is the area of $YZRS$?



Solution 15 *Area* $PQRS$ *is* $64cm^2$. *Area of* $\triangle XSR = \frac{1}{2}(8)(8) = 32cm^2$. *Take* T *to be midpoint of* SR, *then*



Each of the four smaller triangles contained in $\triangle XSR$ *has equal area, which is* $\frac{1}{4}(32) = 8cm^2$. *Since area of* $YZRS$ *is made of three of these triangles, it has an area of* $3 \times 8 = 24cm^2$.